

Soil structure and the saturated hydraulic conductivity of subsoils

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Abstract

The saturated hydraulic conductivity, K_{sat} , was measured on soil samples collected from the plough layer and the subsoil. A range of naturally occurring soil bulk densities was obtained by sampling in different years, with different crops and within and without wheel-tracks, etc. It was found that, for the plough layer, quite good linear relationships exist between the logarithm of K_{sat} and the bulk density. However, for the subsoils, the value of K_{sat} usually lies above the regression line for found for the corresponding plough layer. This “excess” hydraulic conductivity of subsoils is attributed to the presence of biopores, especially root channels. The lower hydraulic conductivity of the plough layer, relative to the subsoil, is attributed to the destruction of these biopores by tillage. A simple model for the separate contributions of soil texture and root channels to the overall value of K_{sat} is presented. It is concluded that subsoil tillage could cause significant reductions in K_{sat} with potentially serious environmental consequences unless it is repeated periodically.

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1. Introduction

The hydraulic conductivity of saturated soil is of great importance for both agricultural production and environmental protection. Saturated hydraulic conductivity, K_{sat} , controls the infiltration of water into soil, especially at long times. Low values of K_{sat} are associated with ponding of water on the soil surface, anaerobic (reducing) soil conditions, run-off, flooding and erosion.

Of special importance is the K_{sat} of the soil layer immediately below the tilled layer. This layer we shall

call the “subsoil”. In many cases, this layer has been compacted by the combined effects of compaction by heavy vehicles and by the pressures exerted by the bottoms of the tillage implements (e.g. ploughs). In Poland, primary tillage is usually done to a depth of 25 cm. The “plough layer” (0–25 cm depth) often has the same particle size distribution as the “subsoil” (>25 cm depth), and therefore their hydraulic properties can be compared directly.

2. Theory

The conductivity of water in soil is analogous to the electrical conductivity of an electrical resistance network. When there are distinctly different modes

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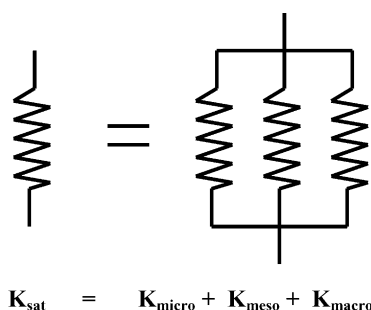


Fig. 1. Electrical resistance analogue of a soil with pathways for conductance of water through micro-, meso- and macro-structural pores. In this case, electric current is the analogue of water flow.

of transport, the soil can be modelled as a simple parallel resistor network as shown in Fig. 1.

When all the samples have the same size, then the conductances are proportional to the conductivities. In this case, the conductivities are additive, and the total conductivity is given by:

$$K = K_{\text{micro}} + K_{\text{meso}} + K_{\text{macro}} \quad (1)$$

The Polish soils considered in this paper have low clay contents, and macro-structural features, such as desiccation cracks, do not usually occur. Therefore, we can assume that $K_{\text{macro}} = 0$, and consider only the first two terms.

Because of the wide range of values of K , we plot graphs of the logarithms (to base 10):

$$\log_{10} K = \log_{10}(K_{\text{micro}} + K_{\text{meso}}) \quad (2)$$

3. Soils and experimental methods

Soil samples were collected from four different sites in Poland. Information about the sites and the soil compositions is given in Table 1. Samples from the

tilled layer were typically collected from the 10–16 cm depth interval, and samples from the subsoil were typically collected from the 30–36 cm depth interval.

Sites A and D are fields located on experiment stations of our research institute (IUNG) whereas, sites B and C are on private, commercial farms. No compaction treatments were applied. Instead, the range of soil densities found had arisen as a consequence of sampling on different dates, the use of different crop rotations and other management practices, etc.

Measurements of saturated hydraulic conductivity, K_{sat} , were made by the falling-head method (Hartge and Horn, 1992). The K_{sat} samples were of 8 cm diameter and 8 cm length. Measurements of dry bulk density were made on samples collected in 100 cm³ stainless steel cylinders.

4. Results

Measured values of the saturated hydraulic conductivity for the four different experimental sites are shown in Fig. 2. Each point on these graphs represents the geometric mean from measurements made on 10 replicate samples for K_{sat} and the arithmetic mean of four replicate samples for bulk density, ρ , collected from a small area (about 1 m²). Geometric means were used for K_{sat} because the values have been shown to be log-normally distributed to within experimental error as was also found by Baker and Bouma (1976). Typical mean values of $\log K_{\text{sat}}$ and of ρ , and of their variation, are given in Table 2. Note that because of the replication, for bulk density the S.E. values are one-half of the S.D. values; whereas, for $\log K_{\text{sat}}$ the S.E. values are about one-third of the S.D. values.

The agricultural topsoils which have been tilled regularly show a linear decrease in the logarithm of

Table 1
Experimental soils

Soil	Location	Position	Texture class	Clay (%) ($<2 \mu\text{m}$)	Silt (%) ($2\text{--}50 \mu\text{m}$)	Sand (%) ($>50 \mu\text{m}$)
A	Grabow	51°20'58"N 21°39'51"E	Sandy loam	3	26	71
B	Huta	51°16'28"N 22°05'51"E	Sandy loam	3	33	64
C	Tomaszow Lubelski	50°35'26"N 23°23'53"E	Silt loam	7	80	13
D	Baborowko	52°35'02"N 16°38'36"E	Sandy loam	8	21	71

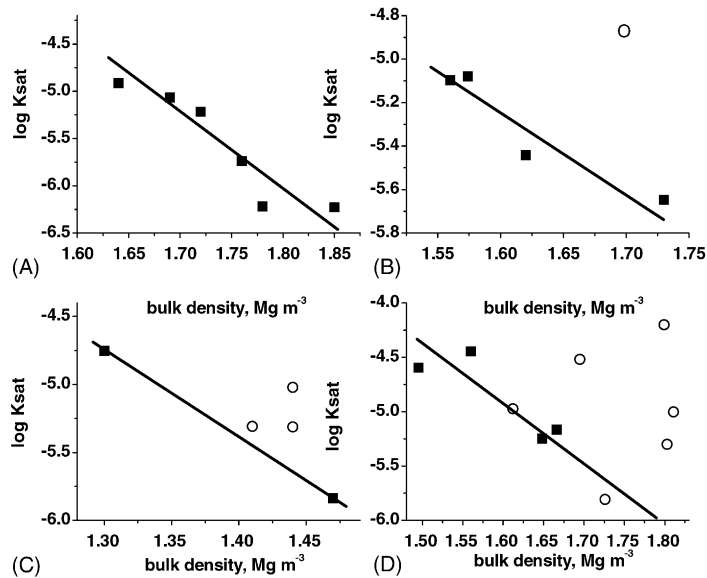


Fig. 2. Measured values of the saturated hydraulic conductivity, K_{sat} , for the four experimental sites. Topsoil measurements are shown as solid squares whereas sub-soil measurements are shown as open circles.

saturated hydraulic conductivity, K_{sat} , with increasing bulk density, ρ . This can be expressed as:

$$\log K_{\text{sat}} = a + b\rho, \quad (3)$$

where a and b take different empirical values for different soils. The regression lines are shown in Fig. 2. The values of the coefficients a and b obtained by regression for the tilled layers of the experimental soils are given in Table 3.

Subsoils that have a similar particle-size distribution, often have values of hydraulic conductivity which are larger than that given by Eq. (3) for the corresponding topsoil. This is illustrated in Fig. 2

where the subsoil values (shown as open circles) mostly lie above the regression lines for the corresponding tilled topsoils. For the Polish sandy soils investigated, we attribute this “excess” hydraulic conductivity to the presence of meso-pores, usually in the form of root channels.

We have investigated this “excess” hydraulic conductivity by subtracting from the measured values the values predicted from Eq. (3) using the coefficients given in Table 3. This gives $K_{\text{excess}} = K_{\text{sat}} - K_{\text{micro}}$. In these calculations, we used values of K_{sat} rather than $\log K_{\text{sat}}$ to be in accordance with Eqs. (1) and (2). The “excess” values of K_{sat} , thus calculated, were

Table 2

Typical mean values of $\log K_{\text{sat}}$ and of bulk density, ρ , and their variation expressed as the standard deviations, S.D., of the values from replicate samples

Soil	Typical value of density (Mg m^{-3})	Typical S.D. of density (Mg m^{-3})	Typical value of log (K_{sat} in m s^{-1})	Typical S.D. of log K_{sat} (K_{sat} in m s^{-1})
A	1.7	0.06	−5.2	0.4
B	1.6	0.04	−5.2	0.2
C	1.35	0.10	−5.2	0.4
D	1.65	0.04	−5.2	0.3

Table 3

Coefficients a and b of Eq. (3) for the tilled layers of the experimental soils

Soil	a	b
A	7.28 (± 2.47)	−7.38 (± 1.42)
B	0.19 (± 1.34)	−3.40 (± 0.83)
C	3.53 (n.a.)	−6.37 (n.a.)
D	3.91 (± 2.17)	−5.53 (± 1.34)
Mean	3.73	−5.67

Values in parenthesis are standard errors. n.a.: not available.

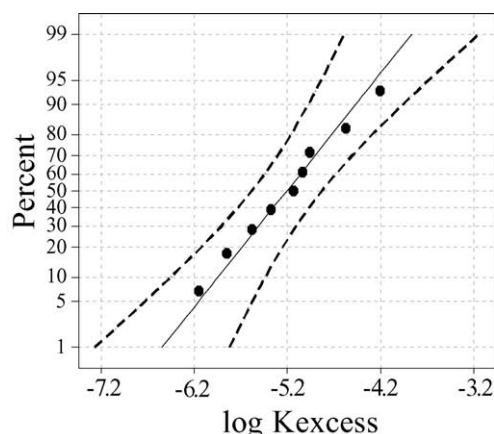


Fig. 3. Normal probability plot of the logarithms (to base 10) of the “excess” values of K_{sat} (m s^{-1}) for the sub-soils.

combined for all the soils B, C and D because of the small number of values available. The distribution of the logarithms of the combined values was fitted to a normal distribution. The resulting probability plot is shown in Fig. 3. This normal distribution has a mean of $\log K_{\text{excess}} = -5.212$ and a standard deviation of $\text{S.D.} = 0.612$. A Shapiro-Wilk normality test shows that these data give $P = 0.993$ and are distributed normally at the 0.05 level (Shapiro and Wilk, 1965).

We can look at the implications of this by adding the micro-structural conductivity and the meso-structural conductivity for a hypothetical, typical soil, in accordance with Eqs. (1) and (2). For the micro-structural conductivity, we can use Eq. (3) with the

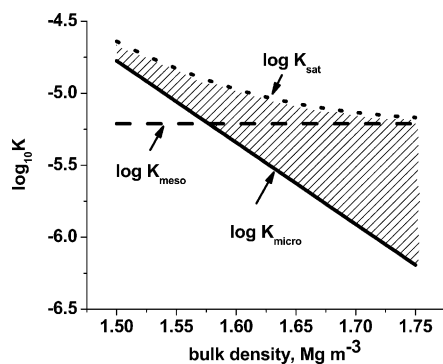


Fig. 4. Hypothetical example of the contributions of micro- and meso-structure of sub-soil to the saturated hydraulic conductivity at different values of bulk density. The area shown shaded represents the hydraulic conductivity that can be lost if the meso-pores (e.g. root channels) are destroyed.

mean coefficients given in Table 3. For the meso-structural conductivity, we can use the mean value from the normal distribution shown in Fig. 3. This produces the graph as shown in Fig. 4.

In the example shown in Fig. 4, at a value of bulk density of about 1.575 mg m^{-3} , the contributions of the micro- and meso-structural pores to the saturated hydraulic conductivity are equal. At densities smaller than this, the micro-structure dominates, whereas at densities larger than this, the meso-structure dominates. Although Fig. 4 is realistic, it must not be forgotten that K_{meso} can vary by a factor of at least 100 as shown in Fig. 3 and 4. This fact illustrates the impossibility of accurately predicting K_{sat} from bulk density alone for structured soils.

5. Conclusions

We conclude that the “excess” saturated hydraulic conductivity of Polish, sandy subsoils is due to the existence of conducting meso-pores, usually in the form of root channels. These meso-pores can contribute significantly to the hydraulic conductivity of subsoils. In some soils, earthworm tunnels could also make a large contribution to K_{sat} , although earthworms are not common in the sandy soils investigated here.

Although the effect of meso-structural features such as root channels is most easily demonstrated for cases where the particle size distributions of the topsoil and the subsoil are the same, the magnitude of the effect can be expected to be similar for cases in which the topsoil and subsoil are different.

A logical consequence of these findings is that subsoil tillage to depths of, for example, 40 cm can destroy the existing subsoil meso-structure. After subsoiling, it has been found that the soil can readily recompact to a similar or greater density to that before subsoiling (e.g. Horn et al., 1998). However, it will recompact without meso-structure and it will have lower values of K_{sat} than it had before deep-tillage (or subsoiling).

Therefore, we can conclude that deep tillage should not be done in cases where subsoil structure could be destroyed and where the soil will be recompacted. In such cases, subsoil tillage could have the severe implications of increasing water ponding, run-off and erosion through the consequent reduction of K_{sat} . The

implications of loss of meso-structure will be more severe in more-highly compacted subsoils.

The presence of meso-structure, such as root channels, means that saturated hydraulic conductivity cannot be predicted adequately from values of soil bulk density alone.

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